Gate-Field-Plated InGaO High-Voltage TFT on a Si Substrate With a Breakdown Voltage of 1348 V

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***Abstract*— In this work, we report back-end-of-line (BEOL)-compatible (< 300 °C) gate-field-plated (GFP) InGaO (IGO) high-voltage thin-film transistors (HVTFTs) on Si substrates. The fabricated GFP IGO HVTFT demonstrated a low subthreshold swing (*SS*) of 95 mV/dec and a positive threshold voltage (V_{TH}) of 0.16 V. For a device with a gate-to-drain offset length of 15 μm, the off-state measurement exhibited a record high breakdown voltage (V_{BD}) of 1348 V, which is ~2.4 times higher than that of the non-FP IGO HVTFT (547 V), corresponding to a Baliga's figure of merit (BFOM) of 14.2 MW/cm^2. According to electric field simulation by technology computer aided design (TCAD), this significant increase in the V_{BD} is because the peak electric field of the device shifts from the edge of the gate to the SiO_2 dielectric of the field plate. In this work, oxide HVTFTs on Si substrates are decisively pushed into kilovoltage application scenarios.**

***Index Terms*— Gate field plate, indium gallium oxide, high-voltage, thin film transistors, BEOL.**

I. Introduction

HIGH-VOLTAGE thin-film transistors (HVTFTs) based on amorphous oxide semiconductors (AOSs) are attractive for large-area power functions because wide bandgaps enable low leakage and high mobility helps lower on-resistance under BEOL-compatible thermal budgets, which have potential for kilovoltage level application [1], [2], [3], [4]. In the BEOL process of semiconductor manufacturing, HV-TFTs are required as the I/O interface between the CMOS core circuit and peripheral high-voltage loads [5], [6]. Si substrates offer cost and size advantages, particularly in terms of compatibility with monolithic 3D ICs and on-die power management [7], [8], [9], where BEOL-compatible HV devices are being increasingly sought to maximize the breakdown voltage (V_{BD}) while minimizing the specific on-state resistance ($R_{on,sp}$). In recent years, various amorphous oxide semiconductors have been explored for use in high-voltage transistors. IGZO transistors have achieved 464 V via split-channel engineering [10]; ITO transistors reached 605 V using a top-gate structure [11]; IWO transistors exceeded 200 V with a dual-gate architecture [12]; and IGO transistors achieved the highest V_{BD} of 638 V by employing atomic-layer-deposited (ALD)-grown IGO channels [13]. These results indicate that IGO offers the greatest potential for high-breakdown-voltage-power device applications among reported amorphous oxide materials because of their high mobility, wide bandgap, and excellent stability. However, the unmanaged high electric field in the gate-drain depletion region of those devices undermined their potential because of premature impact ionization breakdown. Field plate is an effective field management technique because it shifts the position of the peak electric field and has been widely used in wide-bandgap power devices (AlGaN/GaN HEMTs and β-Ga_2O_3 MOSFETs) to increase the off-state breakdown voltage [14], [15], [16], [17], [18], [19]. Although field plates can also be used to adjust the V_{BD} of AOS HV-TFTs by applying different bias voltages on glass substrates, this increases the complexity of device driving, and the on-state current is substantially low [20].

In this work, we experimentally report gate-connected field-plated ALD IGO HVTFTs on Si substrates below 300 °C. The GFP IGO HVTFT achieves a high off-state breakdown voltage of 1348 V and a BFOM of 14.2 MW/cm^2 on Si substrates. TCAD revealed that the GFP reduces the gate edge electric field from 5.9 MV/cm to 3.6 MV/cm, whereas the peak electric field (9.7 MV/cm) shifts to the SiO_2 dielectric below the GFP and flattens the lateral electric field in the drift region.

II. Device Fabrication

The cross-sectional schematics of the top-gate IGO TFTs without and with a GFP are shown in Figs. 1(a) and 1(b), respectively. The key steps for fabricating the GFP IGO HVTFTs are shown in Fig. 1(c). The dashed box highlights the additional GFP flow. Devices were built on SiO_2/Si substrates with a 300 nm SiO_2 isolation layer. After RCA cleaning, a 10 nm IGO channel was deposited by atomic layer deposition (ALD) at 225 °C and patterned into a rectangular mesa by HCl wet etching, followed by e-beam evaporation of Ni/Au (20/40 nm) to form source/drain contacts [1]. A 15 nm HfBeO (HfO_2: BeO = 9:1) gate dielectric was then deposited by ALD (with O_3 as the oxygen precursor), and the dielectric constant of the 15 nm HfBeO dielectric was 11. This design

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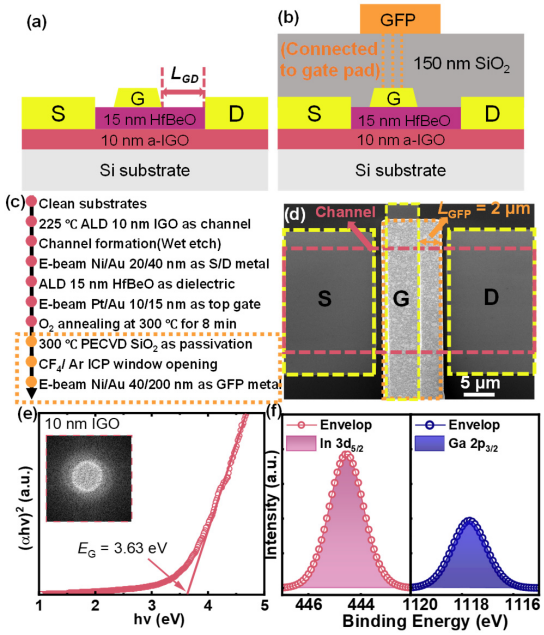


Fig. 1. (a) Device structure schematic of the top-gate TFT for comparison. (b) Schematic cross-section of the GFP IGO HVTFT. (c) Fabrication process of the GFP IGO HVTFTs. (d) SEM image of a fabricated device with L_G/L_{GD} = 5 μm/3 μm and a length of GFP extension of 2 μm. (e) Tauc plot of the IGO films. The fitting of the curve is based on the Tauc model (α: absorption coefficient; h: Planck's constant; ν: frequency of vibration). (f) XPS spectra for In $3d_{5/2}$ and Ga $2p_{3/2}$ core levels of IGO films. Insert (e): fast Fourier transform (FFT) image obtained for amorphous IGO films.

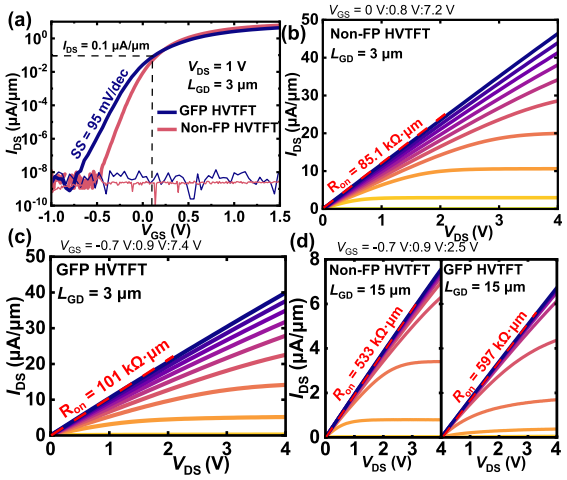


Fig. 2. (a) Experimental transfer curves for the non-FP and GFP IGO HVTFTs. (b) Experimental output curve of a non-FP IGO HVTFT with an L_{GD} of 3 μm. (c) Experimental output curve of the GFP IGO HVTFT with an L_{GD} of 3 μm. (d) Experimental output curves of the non-FP and GFP IGO HVTFTs with L_{GD} = 15 μm.

combines the high dielectric constant of HfO_2 with the wide bandgap energy (~9.4 eV) of BeO [21]. The gate electrode Pt/Au (10/15 nm) was formed by e-beam evaporation. O_2 rapid thermal annealing (RTA) at 300 °C for 8 min was used to set a positive threshold voltage [13]. The nominal device dimensions were $L_G/W_{CH}/L_{GS}/L_{GD}$ = 5/20/1.5/3 μm. A 150 nm SiO_2 passivation layer was grown by plasma-enhanced chemical vapor deposition at 300 °C, which served as the surface passivation and GFP mechanical support, while suppressing surface leakage and thereby contributing to improved breakdown capability [14]. The electrode pad contact windows were opened by low-power CF_4/Ar inductively coupled plasma (ICP). Finally, a Ni/Au (40/200 nm) GFP was deposited by e-beam evaporation and connected to the gate pad, as shown in Fig. 1(b). The DC characteristics were measured at room temperature via an Agilent B1500A semiconductor device analyzer. Off-state breakdown tests were performed on a Cascade Tesla power probe station with a B1505A semiconductor device analyzer. The breakdown measurements were carried out with devices submerged in Fluorinert FC-40 to rule out any air arcing [19], [22]. Fig. 1(d) shows an SEM image of a device with a GFP, showing a length of 2 μm GFP extension toward the drain and precise alignment to the gate finger. The Tauc plot extracted from the optical absorption spectrum is shown in Fig. 1(e), indicating a bandgap of 3.63 eV for the IGO films, yielding an estimated critical breakdown electric field of approximately 4 MV/cm for the IGO films [23]. The insert of Fig. 1(e) shows an FFT image of the IGO films, which indicates that the IGO channel is amorphous. The X-ray photoelectron spectroscopy (XPS) spectra of the In $3d_{3/2}$ and Ga $2p_{5/2}$ core levels of the IGO films are shown in Fig. 1(f), indicating that the atomic ratio of In to Ga was 74:26.

III. Results and Discussion

The transfer curves of the GFP IGO HVTFT device and the non-FP device at a V_{DS} of 1 V are compared in Fig. 2(a). The GFP IGO HVTFT device shows a leftward turn-on ΔV_{on} shift of -0.3 V with a slightly reduced on-current and a modestly degraded SS, which correlated with hydrogen diffusion into the IGO channel during SiO_2 deposition in top-gate devices [24]. Even so, the GFP IGO HVTFT maintains strong gate control: in the off state, the drain current is gate leakage limited at ~ 5.1×10^{-9} μA/μm, yielding an excellent I_{ON}/I_{OFF} of 10^9, an *SS* of 95 mV/dec, and a V_{TH} of 0.16 V (defined at an I_{DS} of 0.1 μA/μm). A low SS of 95 mV/dec indicates a low interface trap density D_{it} value of 2.3×10^{12} $cm^{-2}eV^{-1}$ when $SS = 60$ mV/dec $(1 + qD_{it}/C_{ox})$. The extremely low I_{off} could be attributed to the gate-to-drain lenth (L_{GD}) eliminating the leakage paths within the overlapping area between the gate and source/drain [25]. Comparisons of the output characteristics with an L_{GD} of 3 μm, the non-FP IGO HVTFT reaches a maximum saturation drain current ($I_{DS,sat}$) of 47.0 μA/μm, whereas the GFP device yields *an* $I_{DS,sat}$ of 39.6 μA/μm at a V_{DS} of 4 V. The reduction in $I_{DS,sat}$ and transconductance (G_m) was attributed to the extension of the GFP, which led to an increase in the gate capacitance (C_{GD}) [26]. The output curves of the non-FP IGO HVTFT and the GFP IGO HVTFT with an L_{GD} of 15 μm are compared in Fig. 2(d). The enlarged offset region reduces the current to $I_{DS,sat}$ values of 7.5 μA/μm and 6.7 μA/μm. A specific on-resistance ($R_{on,sp} = R_{on} \cdot W \cdot L_{SD}$)$R_{on,sp}$ of 114.7 mΩ·cm^2 and 128.4 mΩ·cm^2 can be extracted from the output curve at a V_{GS} of 2.5 V.

The three-terminal breakdown characteristics are shown in Figs. 3(a) and 3(b). During the breakdown measurement, the device was maintained in the normal-off state [27]. The abrupt

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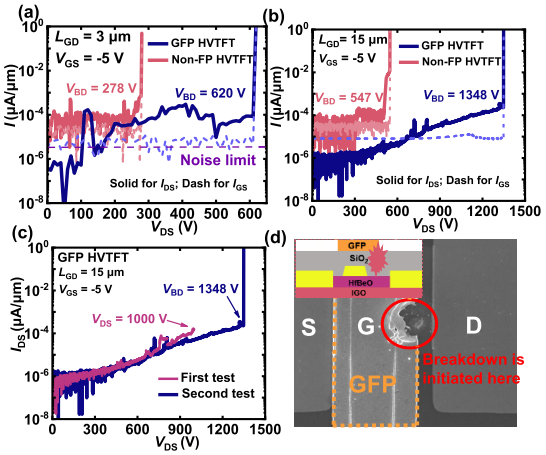


Fig. 3. Experimental off-state breakdown characteristics of the non-FP and GFP IGO HVTFTs (a) with an L_{GD} of 3 μm (b) and an L_{GD} of 15 μm. (c) The kV-level high-voltage stress test for the GFP IGO HVTFT was conducted under the following conditions: the first test: V_{DS} = 1000 V; the second test: the device failed completely. (d) SEM image of a post-breakdown device, and the position where breakdown is initiated is indicated.

exponential increase in both the I_{DS} and I_{GS} indicates that the device has undergone complete breakdown. Comparisons were conducted among devices with L_{GD} of 3 μm and 15 μm. Specifically, the GFP IGO HVTFT breakdown voltage reached 1348 V, which was 2.4 times greater than that of the non-FP IGO HVTFT (547 V). This significant difference clearly illustrates the alleviating effect of the GFP on the internal electric field of the device. On the basis of the calculation formula (BFOM = $V_{BD}^2/R_{on,sp}$), the GFP IGO HVTFT achieves a BFOM of 14.2 MW/cm^2, which was 5.5× greater than that of the non-FP IGO HVTFT (2.6 MW/cm^2). Notably, the GFP IGO HVTFT shows an off-state I_{DS} that is ~ two orders of magnitude lower because the GFP moderates the drain-side gate-edge field, suppresses gate leakage and impacts ionization-driven carrier multiplication before breakdown [28]. To assess the robustness under high-stress conditions, the device was subjected to two tests (Fig. 3(c)). In the first test, a stress of V_{DS} of 1000 V was applied, and the device withstood this kilovoltage stress without failure. In the second test, the stress gradually increased until the device eventually failed. The trend of the I_{DS} changes was essentially consistent in the two tests, which indicates that the GFP IGO HVTFT can repeatedly withstand kilovoltage-level voltages, a property of great significance for the practical application of the device. An SEM image of the device after breakdown is shown in Fig. 3(d), indicating that the breakdown most likely initiates in the SiO_2 passivation layer at the edge of the GFP. At this stage, no breakdown is observed at the electrodes.

To understand the electric field profiles and breakdown location within the device, the structures were simulated using a Sentaurus TCAD under the experimental breakdown conditions, as shown in Figs. 4(a) and 4(b). Compared with the non-FP HVTFT (Figs. 4(c) and 4(d)), the peak electric field shifts from the gate dielectric to SiO_2 passivation, which is simulated to be 9.7 MV/cm at the FP edge. Moreover, the electric field at the gate edge is reduced to 3.6 MV/cm. Combined with the post breakdown SEM observation in

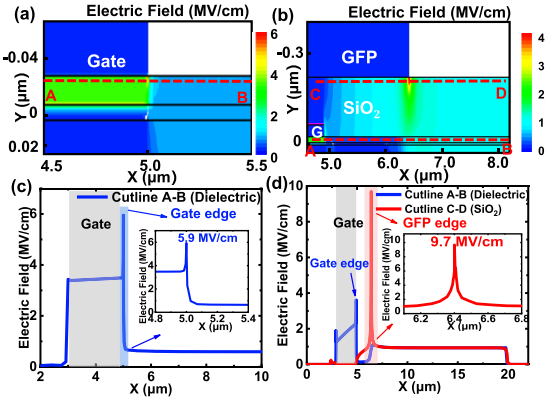


Fig. 4. Simulation of the 2D electric field (a) non-FP IGO HVTFT and (b) GFP IGO HVTFT for L_{GD} = 15 μm. (c) Lateral electric field profile under the gate along cutoff A–B for the non-FP IGO HVTFT. (d) Lateral electric field profile under the gate and the FP along cutlines A-B and C-D under the experimental breakdown condition of V_{BD} = 1348 V for the GFP IGO HVTFT.

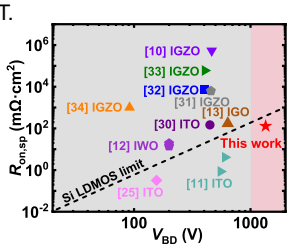


Fig. 5. Benchmark of $R_{on,sp}$ versus V_{BD} of the GFP IGO HVTFT in this work, previously reported AOS HVTFTs on a Si substrate.

Fig. 3(d), it is therefore possible that breakdown initiates during SiO_2 passivation and that gate-related tunneling leakage also serves as a dominant breakdown pathway for the GFP IGO HVTFT with a large L_{GD} [29].

The performance of the GFP IGO HVTFT is benchmarked against some representative AOS HVTFTs on Si substrates in the plot of $R_{on,sp}$ versus V_{BD} in Fig. 5 [10], [11], [12], [13], [25], [30], [31], [32], [33], [34]. In this work, the GFP IGO HVTFT exhibits a V_{BD} of 1348 V, representing a 2.1× improvement over the record (638 V) and entering the kilovoltage application for AOS HVTFTs. These results indicate that the GFP is an effective way to push the AOS HVTFTs beyond the theoretical limit of the single-RESURF Si LDMOS.

IV. Conclusion

In conclusion, an IGO HVTFT with a gate-connected field plate is fabricated on Si substrates to shift the peak electric field from the dielectric layer to the SiO_2 passivation layer. The GFP IGO HVTFT achieves an $R_{on,sp}$ of 128.4 mΩ·cm^2, a record breakdown voltage of 1348 V and a BFOM of 14.2 MW/cm^2, surpassing the theoretical limit of the single-RESURF Si LDMOS. The device is fabricated using a BEOL-compatible thin-film process, demonstrating significant potential for future applications in monolithic three-dimensional integration.

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